

1 VQ-VAE: Discrete Latent Codes

Vector-Quantized VAEs map images to discrete latent tokens. An encoder converts a 256×256 image into a 32×32 grid of discrete codes; a decoder reconstructs the image from these codes. The training objective includes:

$$\log p(x \mid q(z_e(x))) + \|\text{sg}[z_e(x)] - e\|_2^2 + \beta \|z_e(x) - \text{sg}[e]\|_2^2.$$

(See slide 3 for diagram.)

2 Diffusion Models: Intuition

Diffusion models gradually corrupt data with Gaussian noise. A neural network learns to reverse this noising process by predicting the noise added at each step. After training, sampling proceeds by starting from pure noise and iteratively denoising. (Visual examples on slide 4.)

3 Formal Diffusion Process

A forward noising Markov chain adds noise over T steps:

$$q(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I),$$

leading to x_T approaching an isotropic Gaussian. The reverse process is parameterized by a neural network:

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma_t^2 I).$$

(See slide 5.)

4 Training Objective

Training maximizes a variational lower bound on $\log p_\theta(x_0)$. When forward variances β_t are fixed, the posterior $q(x_{t-1} \mid x_t, x_0)$ has no learnable parameters. Using the reparameterization:

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon,$$

the network is trained to predict the noise ϵ . The simplified loss (Ho et al. 2021) is:

$$\mathbb{E}_{t,x_0,\epsilon} \|\epsilon - \epsilon_\theta(x_t, t)\|_2^2.$$

(Slide 6.)

5 Sampling and Architecture

Sampling applies the learned reverse transitions:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right) + \sigma_t z.$$

Most implementations use a U-Net because input and output shapes match. (Slides 7–9.)

6 Summary of Key Properties

- Diffusion is parameterized as a Markov chain; transitions are Gaussian.
- Forward variances are fixed or learned; reverse means are learned.
- Training optimizes a variational upper bound on $-\log p(x)$.
- KL terms simplify due to Gaussian and Markov assumptions.
- Predicting noise directly yields the most stable training.
- U-Net-like architectures fit naturally because x_t and x_{t-1} share dimensionality.

(See slide 10.)

7 Cold Diffusion

Cold Diffusion replaces Gaussian noise with arbitrary image degradations (e.g., blur, pixelation, masks). The reverse model learns to invert these transformations without stochastic noise. (Slide 11.)

8 Stable Diffusion

Two extensions:

- (a) Perform diffusion in a latent space (via an autoencoder or VQ-VAE), offering major speed benefits.
- (b) Condition the reverse process on text embeddings, using cross-attention between text and image latents.

(Slides 12–13.)